Samarth Kumar Samal

Data Scientist | Clinical Data Science Researcher | Machine Learning

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Research Interests

Clinical Data Science
Healthcare Machine Learning
Biomedical Signal Processing
Gait and Parkinson's Disease
Electronic Health Records
Clinical Phenotyping

Technical Skills

Programming
Python, R, SQL, C++, Rust

Machine Learning
Scikit-learn, classification, clustering, feature engineering, cross-validation, model evaluation

Statistics and Data Science
Pandas, NumPy, SciPy, statistical analysis, missing-data handling

Clinical Data Science
eICU, MIMIC-III, EHR analytics, clinical phenotyping, ICD/CPT modeling

Biomedical Signal Processing
Six-axis IMU, time-domain analysis, frequency-domain analysis, Welch PSD, spectrograms, gait analysis

Clinical NLP and Deep Learning
BioClinicalBERT, Longformer, BiLSTM, multi-label classification

Visualization
Matplotlib, Seaborn, Plotly, Streamlit

Development Tools
Git, GitHub, Docker, Google Cloud Platform, Linux, JupyterLab, VS Code, Jira

Databases
MySQL, Microsoft SQL Server

Languages

English – Professional Proficiency
Hindi – Professional Proficiency
Odia – Professional Proficiency

Professional Profile

Graduate Data Science researcher at the University of Minnesota Twin Cities with two years of software engineering experience at Dell Technologies. Experienced in clinical data science, healthcare machine learning, biomedical signal processing, electronic health record analytics, and software engineering. Current research focuses on unsupervised discovery of ICU patient subtypes and IMU-based gait analysis for Parkinson's disease. Skilled in developing reproducible Python workflows for clinical data preprocessing, statistical analysis, clustering, signal processing, natural language processing, and model evaluation.

Education

University of Minnesota Twin Cities **2025–Present**

Master of Science in Data Science; Minor in Computer Science *Minneapolis, Minnesota*
GPA: 3.61/4.00

Odisha University of Technology and Research **2018–2022**

Bachelor of Technology in Electrical Engineering *Bhubaneswar, India*
GPA: 3.83/4.00

Research Experience

Volunteer Researcher – Parkinson's Disease Research **May 2026–Present**

Neuromodulation Center, Department of Neurology *Minneapolis, Minnesota*
Principal Investigator: Prof. Biswaranjan Mohanty

- Analyze and preprocess six-axis inertial measurement unit (IMU) data comprising three-axis accelerometer and three-axis gyroscope signals from healthy participants and individuals with Parkinson's disease.
- Develop a reproducible workflow for continuous, unlabeled IMU recordings to identify resting, walking, and turning segments, detect gait cycles, and classify healthy and Parkinsonian gait patterns.
- Extract clinically relevant gait measures including stride length, step length, cadence, swing time, stance time, and walking speed.
- Perform time-domain and frequency-domain analysis, including power spectral density and spectrogram analysis, together with feature extraction, cross-validation, and visualization.

Capstone Researcher – ICU Clinical Subtype Discovery **May 2026–Present**

Department of Surgery, Division of Computational Health Sciences *Minneapolis, Minnesota*
Principal Investigator: Prof. Jocelyn Chi

- Build a reproducible analytic dataset from the eICU Collaborative Research Database using first-24-hour demographic, laboratory, vital-sign, physiologic, and illness-severity variables for unsupervised clinical phenotype discovery.
- Perform cohort selection, data-quality assessment, missing-value handling, repeated-measurement aggregation, feature engineering, and scaling before clustering analysis.
- Evaluate K-Means and alternative clustering solutions using within-cluster sum of squares, silhouette score, cluster size, stability, consistency, clinical interpretability, and clinical plausibility.
- Characterize ICU subtypes using clinical profiles, APACHE IVa severity, ICU and hospital mortality, and length of stay to assess clinical relevance.

Certifications

- Advanced Certification in Data Science and Artificial Intelligence
- Artificial Intelligence
- Machine Learning
- Introduction to Programming Using Python
- Microsoft SQL Server

Selected Coursework

- Machine Learning
- Deep Learning
- Data Mining
- Natural Language Processing
- Applied Multivariate Methods
- Operating Systems

Leadership and Campus Involvement

Alumni Coordinator
Odisha University of Technology and Research
2019–2022

- Organized annual alumni events in physical and virtual formats.
- Connected alumni and students through workshops, webinars, and career discussions.

E-Cell Head
Odisha University of Technology and Research
2020–2021

- Guided junior students on Arduino-based projects, including line-following robots.

Professional Experience

Software Engineer I **August 2022–July 2024**

Dell R&D Center India *Bengaluru, India*

- Developed and maintained Dell Update Package and USB Network Interface Card workflows across iDRAC, USBNIC, PSU, CPLD, and device-driver components for Windows, WinPE, and Linux enterprise server environments.
- Independently resolved more than 50 Jira issues and contributed to over 100 additional technical investigations through log analysis, source-code debugging, system diagnostics, testing, and fix validation.
- Led migration of the USBNIC library from legacy C++ components to Rust, improving memory safety, maintainability, and long-term reliability of firmware deployment functionality.
- Integrated and validated USBNIC firmware-update components within enterprise deployment workflows to strengthen update consistency and system reliability.
- Provisioned and configured more than 50 virtual machines and 10 physical servers for development, debugging, testing, and validation across multiple operating-system environments.
- Collaborated with engineering, Quality Assurance, DevOps, Cloud Server, and Operating System Management teams across India and the United States to troubleshoot platform issues, coordinate requirements, and improve release readiness.

Selected Technical Projects

COVID-19 Forecasting and Analytics Web Application **December 2025**

GitHub | Live Demo

- Built and deployed an interactive COVID-19 analytics platform using Python, Pandas, Streamlit, Plotly, and Prophet, processing 161,568 records across 198 countries and 816 reporting dates.
- Engineered a reusable data-processing workflow to validate dates, handle missing and non-numeric values, and analyze confirmed cases, recoveries, deaths, daily changes, and epidemiological ratios.
- Developed country-level forecasting for configurable 1–5 year horizons using Prophet with weekly and yearly seasonality and 95% prediction intervals.
- Designed interactive KPI summaries, choropleth maps, country comparisons, ranking views, forecast visualizations, and downloadable filtered datasets.

Temporal Clinical Reasoning for Medical Coding **July 2026**

GitHub | Project Page

- Developed an admission-level multi-label clinical NLP framework using MIMIC-III to jointly predict ICD-10 diagnosis and CPT procedure codes from chronologically ordered electronic health record notes.
- Encoded nursing, physician, and radiology documentation with BioClinicalBERT and Longformer and applied a bidirectional LSTM to capture temporal patient progression across the first five clinical events of each admission.
- Constructed preprocessing for adult admission selection, timestamp ordering, text normalization, ICD-9 to ICD-10 mapping, CPT extraction, and patient-level train, validation, and test splitting using an 80/10/10 ratio.
- Evaluated BioClinicalBERT, Longformer, and temporal BERT-LSTM architectures using Micro-F1, Macro-F1, and Precision@5; Longformer achieved a 0.5925 joint score, while the temporal model reached 0.6096 CPT Micro-F1 and 0.3971 CPT Macro-F1.